

✓ Assignment 1.1.

Introduction to Data Cleaning

Objectives:

- Analyze and interpret the structure and content of raw datasets.
- Apply foundational data preprocessing techniques, such as handling missing values and data type conversions.
- Construct a basic data cleaning pipeline to process real-world data efficiently and with reproducibility.

✓ Procedures

Download the dataset for this activity [here](#)

1. Load the dataset then show the first 5 and last 5 datapoints

```
import pandas as pd

df = pd.read_csv("/content/drive/MyDrive/vda/dirty_cafe_sales.csv")
```

```
from google.colab import drive
drive.mount('/content/drive')
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call

```
df.head()
```

	Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
0	TXN_1961373	Coffee	2	2.0	4.0	Credit Card	Takeaway	2022-01-01
1	TXN_4977031	Cake	4	3.0	12.0	Cash	In-store	2022-01-01
2	TXN_4271903	Cookie	4	1.0	ERROR	Credit Card	In-store	2022-01-01
3	TXN_7004554	Cake	2	5.0	10.0	Credit Card	In-store	2022-01-01

3 TXN_/034554 Salad 2 5.0 10.0 UNKNOWN UNKNOWN 202

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df.tail()

	Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction
9995	TXN_7672686	Coffee	2	2.0	4.0	NaN	UNKNOWN	202
9996	TXN_9659401	NaN	3	NaN	3.0	Digital Wallet	NaN	202
9997	TXN_5255387	Coffee	4	2.0	8.0	Digital Wallet	NaN	202

QUESTION: What Column has numerical values?

Enter your answer here: **Quantity, Price Per Unit, and Total Spent.**

2. Check if there are any missing values in the dataset

df.describe()

	Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction
count	10000	9667	9862	9821	9827	7421	6735	
unique	10000	10	7	8	19	5	4	
top	TXN_9226047	Juice	5	3.0	6.0	Digital Wallet	Takeaway	UNKNOWN

df.isna().sum()

	0
Transaction ID	0
Item	333
Quantity	138
Price Per Unit	179
Total Spent	173

Payment Method 2579

Location 3265

Transaction Date 159

dtype: int64

QUESTION: How many columns had missing values? Which column had the highest number of missing values?

Enter your answer here: 7 columns had missing values, and the highest number missing values is "Location".

3. Remove unnecessary columns and make sure numerical columns only contain numbers.

```
df.columns
```

```
Index(['Transaction ID', 'Item', 'Quantity', 'Price Per Unit', 'Total Spent',  
      'Payment Method', 'Location', 'Transaction Date'],  
      dtype='object')
```

```
df = df.drop(columns=['Transaction ID', 'Payment Method', 'Location', 'Trans  
df.head()
```

	Item	Quantity	Price Per Unit	Total Spent
0	Coffee	2	2.0	4.0
1	Cake	4	3.0	12.0
2	Cookie	4	1.0	ERROR
3	Salad	2	5.0	10.0
4	Coffee	2	2.0	4.0

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```
df.dtypes ## Checking the datatype for each column
```

```
0
```

Item	object
Quantity	object
Price Per Unit	object
Total Spent	object

dtype: object

```
df["Quantity"] = pd.to_numeric(df["Quantity"], errors='coerce')
df["Price Per Unit"] = pd.to_numeric(df["Price Per Unit"], errors='coerce')
df["Total Spent"] = pd.to_numeric(df["Total Spent"], errors='coerce')
```

```
df.head()
```

	Item	Quantity	Price Per Unit	Total Spent
0	Coffee	2.0	2.0	4.0
1	Cake	4.0	3.0	12.0
2	Cookie	4.0	1.0	NaN
3	Salad	2.0	5.0	10.0
4	Coffee	2.0	2.0	4.0

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```
df.dtypes ## Checking the datatype for each column
```

	0
Item	object
Quantity	float64
Price Per Unit	float64
Total Spent	float64

dtype: object

QUESTION: What happened to the object columns?

Enter your answer here: The number columns are converted from **object** to **float 64**, while the Error values turn into **NaN (Not a Number)**.

4. Remove missing data to ensure all that the final dataset is clean

```
df.isna().sum()
```

	0
Item	0
Quantity	0
Price Per Unit	0
Total Spent	0

dtype: int64

```
df = df.dropna()
```

```
df.isna().sum()
```

	0
Item	0
Quantity	0
Price Per Unit	0
Total Spent	0

dtype: int64

```
df.head()
```

	Item	Quantity	Price Per Unit	Total Spent
0	Coffee	2.0	2.0	4.0
1	Cake	4.0	3.0	12.0
3	Salad	2.0	5.0	10.0
4	Coffee	2.0	2.0	4.0
5	Smoothie	5.0	4.0	20.0



Next steps:

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EXPLAIN the process of data cleaning done in the procedure.

Enter your answer here:

1. Check rows, data types, and missing values to find the number of missing values for each column by using the code `df.isna().sum()`.
2. Removed unneeded columns using the code `df.isna().sum()`.
3. Lastly, run the first code to check and confirm if all missing values were removed successfully.

Start coding or [generate](#) with AI.

✓ Supplementary Activities

Download the dataset for supplementary activity [here](#)

1. Load the dataset and show the first 5 and last 5 datapoints
2. Check if there are any missing values in the dataset.
3. The df must only have Name, Subscribers, Category, and Country. Remove all other columns.
4. Convert Subscribers to type **int**
5. Ensure that there are no rows with missing values. (UNKNOWN is counted as a missing value)

```
# 1. Load the dataset and show the first 5 and last 5 datapoints
import pandas as pd

df = pd.read_csv("/content/drive/MyDrive/vda/youtube_subscribers_data.csv")
```

```
df.isna().sum() #first 5 data
```

	0
Name	0
Brand channel	0
Subscribers (millions)	0
Primary language	0
Category	0
Country	0

dtype: int64

df.tail() #last 5 of data

	Name	Brand channel	Subscribers (millions)	Primary language	Category	Country
45	ARY Digital HD	Yes	59.7	Urdu	Entertainment	Pakistan
46	Fede Vigevani	No	59.4	UNKNOWN	Entertainment	Mexico
47	LooLoo Kids	UNKNOWN	59.3	English	Music	Romania
48	Billion Surprise Toys	Yes	UNKNOWN	English	Entertainment	United States

2. Check if there are any missing values in the dataset.

df.isna().sum()

	0
Name	0
Brand channel	0
Subscribers (millions)	0
Primary language	0
Category	0
Country	0

dtype: int64

3. Select only the required columns and rename 'Subscribers (millions)' to

df = df[['Name', 'Subscribers (millions)', 'Category', 'Country']]

df.head()

	Name	Subscribers (millions)	Category	Country
0	MrBeast	335	Entertainment	United States
1	T-Series	280	Music	India
2	Cocomelon - Nursery Rhymes	UNKNOWN	Education	United States
3	SET India	UNKNOWN	Entertainment	India

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```
# 4. Convert to UNKNOWN to NaN
```

```
df['Subscribers (millions)'] = pd.to_numeric(df['Subscribers (millions)'], e  
df = df.dropna() # remove the NaNs  
df['Subscribers (millions)'] = df['Subscribers (millions)'].astype(int) # co  
df = df.rename(columns={'Subscribers (millions)': 'Subscribers'}) #rename su  
df.head()
```

	Name	Subscribers	Category	Country	
0	MrBeast	335	Entertainment	United States	
1	T-Series	280	Music	India	
4	Vlad and Niki	129	Entertainment	Russia	
5	Kids Diana Show	128	Entertainment	United States	Ukraine
6	Like Nastya	123	Entertainment	United States	Ukraine

Next steps:

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✓ Conclusion

In this activity, I learned how to inspect raw data, remove columns that are not needed, fix wrong data types, and clean up missing information.

Good job on finishing the activity! Kindly submit your accomplished google colab link and a PDF file of this activity.